

Auto Ticket Classification using ServiceNow Flow Designer - Student Project

Project Background

For this student project, I considered a school IT helpdesk that receives many incident requests from students and teachers. Common examples are Wi-Fi issues, projector failures, password problems, and slow computers. At present, the staff has to read each request and choose a category manually, so the process takes extra time.

I used ServiceNow Flow Designer to automate the classification. The flow checks keywords in the short description and description fields and then fills the suitable Category and Subcategory.

Main Objectives

- Automatically classify the incident when it is created.
- Reduce manual work for the IT staff.
- Make ticket routing more consistent.
- Keep the solution no-code and easy to maintain.

PHASE 1: Requirement Analysis and Planning

I first listed the common support requests handled by the school IT helpdesk. These include network connectivity, hardware failures, account access problems, and system performance issues.

The staff normally checks each ticket manually to:

- Identify what type of issue was reported.
- Choose the correct category and subcategory.
- Inform the caller that the ticket has been created.

Because these checks are manual, the process can take time and may lead to incorrect classification when the number of requests increases.

Activity 1: Business Use Case

Requirements I used

The system should be able to:

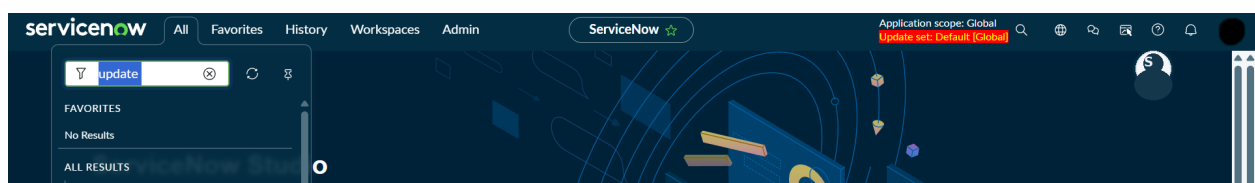
1. Automatically classify IT tickets from the issue description.
2. Assign both Category and Subcategory without manual intervention.
3. Support dependent choices between Category and Subcategory.
4. Send an automatic email to the caller after ticket creation.
5. Store ticket details in a structured format.
6. Be easy to maintain and extend later.

Activity 2: Creating the Update Set

Steps I followed:

- Go to All menus → Search for Update set.
- Under System Update Sets → select Local Update Sets.
- Click New at the top right of the page.
- Name: Project Update Set.
- State: In progress.
- Application: Global.
- Save the form.
- Under Related Links, click Make This My Current Set.
- The created update set will then appear as Current.

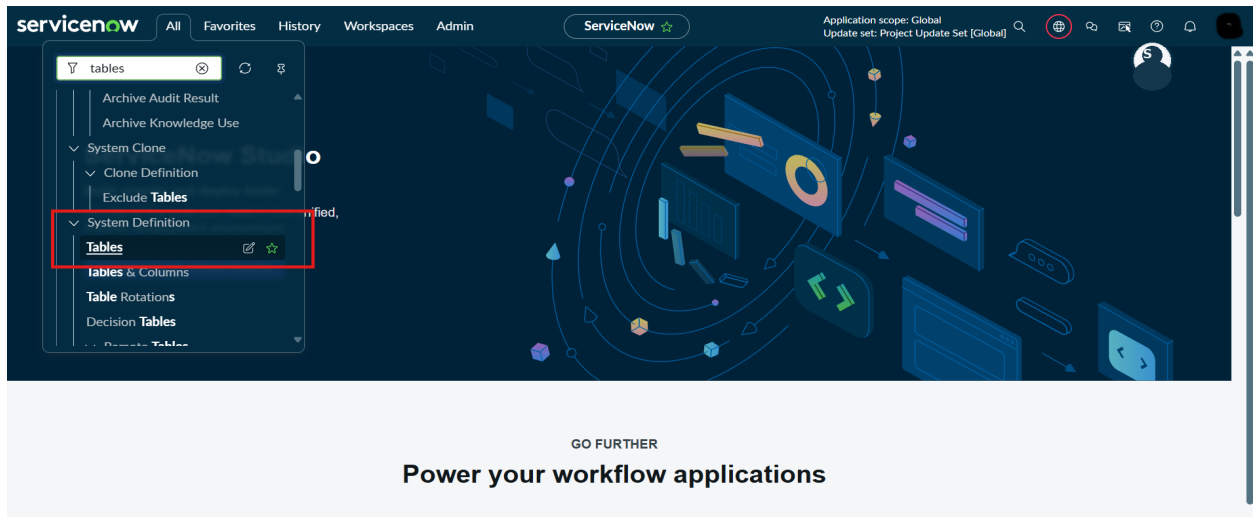
Activity 3: Navigation shown through screenshots



Activity 1: Creating the Custom Table for Ticket Records

Steps I followed to create the table

- Open All menus → Search for Tables.
- Under System Definition → click Tables.
- Click New at the top right corner.
- Label: Incident WorkFlow.
- Uncheck Create module.
- Save the form.
- Open the Controls section and enable Auto Number.
- Save the form again.



Tables

Updated

Search

Actions on selected rows...

New

All > Update name is not empty

Label	Name	Extends table	Extensible	Updated
Search	Search	Search	Search	Search
Logged in User	v_user_session	(empty)	false	2007-07-13 16:15:23
System Plugin	v_plugin	(empty)	false	2007-10-09 12:10:05
Active Transactions	v_transaction	(empty)	false	2008-05-23 17:30:17
Node Log	v_logfiles	(empty)	false	2009-05-14 17:05:41
Customer Uploads	v_customer_uploads	(empty)	false	2009-05-14 18:38:13
MySQL Process List	v_mysql_proclist	(empty)	false	2012-02-25 19:47:02
MySQL Status	v_mysql_status	(empty)	false	2012-02-25 19:47:08
MySQL Variables	v_mysql_variables	(empty)	false	2012-02-25 19:47:12
IOStats	v_iostats	(empty)	false	2012-02-27 17:13:51
Cluster Nodes	v_cluster_nodes	(empty)	false	2012-02-27 23:48:26
SQL Debug	v_sql_debug	(empty)	false	2012-02-28 16:40:55
Workflow Validation Report	v_wf_validation_report	(empty)	false	2012-10-01 10:43:05
Hyper-V Resource Pool Component	cmdb_ci_hyper_v_rpool_comp	Hyper-V Object	false	2012-10-30 08:07:25
Application Menu	sys_app_application	Application File	false	2012-10-30 13:59:18
Menu Category	sys_app_category	Application File	false	2012-10-30 13:59:44
Scriptable Object	v_scriptable_object	(empty)	false	2012-11-09 15:15:23
Model Substitution	cmdb_ci_model_substitute	(empty)	false	2012-12-28 12:28:12

1 to 20 of 5,827

Table
New record

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Incident Workflow
* Name u_incident_workflow

Extends table

Application Global

Create module

Remote Table

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Insert a new row...					

Submit Cancel

Related Links

Table
Incident Workflow

Delete Update Delete All Records

Columns Controls Application Access

Extensible
Live feed

Use auto-numbering to define a sequential identifying code made up of a prefix, a base number and a padding value to ensure a consistent format

Auto-number

Prefix INC

Number 500

Number of digits 5

Security Rules (ACLs) are required if anyone other than an administrator needs to work with this table. Creating default security rules will grant full access to this table to anyone with the user role you specify.

Create access controls

* User role u_incident_workflow_user

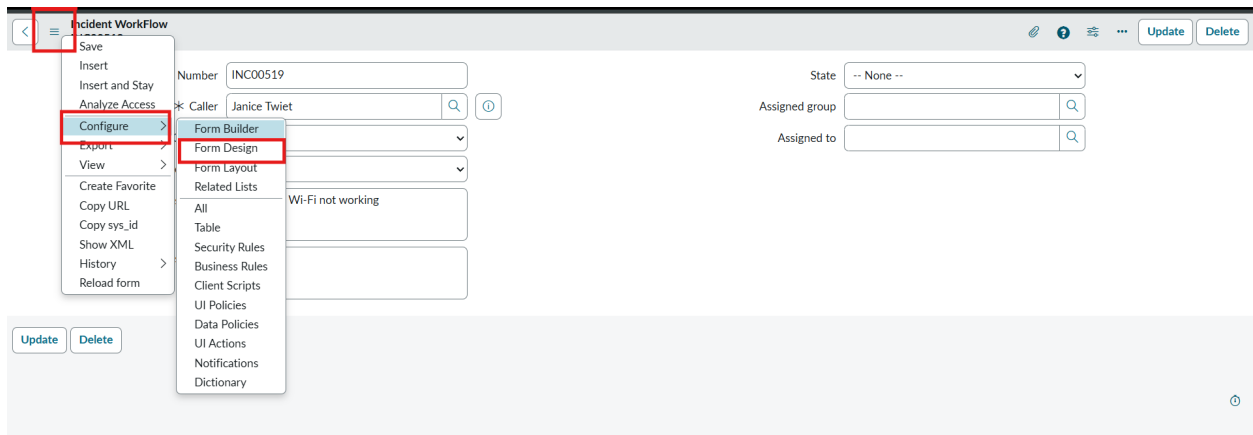
Delete Update Delete All Records

Related Links

[Form Builder](#)

[Define Form](#)

- Save the form and, under Related Links, click Show Form.
- Open Configure from the context menu and then open Form Designer.
- Create the required fields.



Activity 2: Creating Fields and Setting Data Types

In this milestone, I created the fields required for the School IT Ticket table. I selected the data type, choice values, and reference fields so the information is stored consistently.

Field Label	Type	Reference / Choice
Number	Auto Number	—
Caller	Reference	sys_user
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	—
Description	String	—
State	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

- After creating the fields with the required types, add the choices from Form Designer.
- The choice-list fields are Category, Subcategory, and State.
- Add the required choices to these fields.
- Save the form using the Save button at the top right.
- Open the form and reload it to verify the choices.

The screenshot shows the 'Form Design' window for 'Incident Workflow [u_incident_workflow]'. The 'Field Types' panel on the left lists various field types, with 'Choice' highlighted. The 'Properties' dialog for the 'Category' field is open, showing a list of choices. The choices are:

Network	network
Hardware	hardware
Access	access
Performance	performance

The screenshot shows the 'Form Design' window for 'Incident Workflow [u_incident_workflow]'. The 'Field Types' panel on the left lists various field types, with 'Choice' highlighted. The 'Properties' dialog for the 'Subcategory' field is open, showing a list of choices. The choices are:

Wi-Fi	wi-fi
Projector	projector
Forgot Password	forgot password
Slow Computer	slow computer

Activity 3: Linking Category with Subcategory

Here I made the Subcategory field dependent on Category. This means the available subcategory choices change according to the category selected by the user.

Steps I followed

- Open the created table form and right-click the Subcategory field.
- Click Configure Dictionary.
- Under Related Links, open Advanced view.
- Find the Dependent Field section.
- Enable Use dependent field.
- Select Category in the Dependent on field.
- Save the form.

This screenshot shows the 'Incident WorkFlow' 'New record' form. The 'Subcategory' field is highlighted with a red box, and its context menu is open. The menu options are: Configure Label, Configure Dictionary (highlighted with a red box), Configure Styles, Show Security Rules, Configure Choices, Show Choice List, and Show 'u_subcategory'. Other fields visible include Number (INC00509), State (-- None --), Caller, Assigned group, and Assigned to.

This screenshot shows the 'Dictionary Entry Subcategory' configuration page in 'Advanced' view. The 'Table' is set to 'Incident WorkFlow [u_incident_workflow]' and the 'Type' is 'Choice'. The 'Column label' is 'Subcategory' and the 'Column name' is 'u_subcategory'. The 'Application' is 'Global' and 'Active' is checked. Under the 'Dependent Field' tab, 'Use dependent field' is checked, and 'Dependent on field' is set to 'Category'. The 'Choice List Specification' tabs at the top are 'Choice List Specification', 'Dependent Field' (selected), 'Calculated Value', and 'Default Value'.

- Scroll down to the Choice section.
- Add the dependent value for each subcategory using the backend name of the related Category.
- Wi-Fi → Network.
- Projector → Hardware.
- Forgot password → Access.
- Slow computer → Performance.
- Save the form.
- The form will then show the matching subcategory choices based on the selected category.

Dictionary Entry
Subcategory View: Advanced*

Specify a field whose value is used to determine choice list options for this field. [More Info](#)

Use dependent field ☒

Dependent on field Category

Create Choice List Delete Column Update

Related Links
[Show Table](#)
[Run Point Scan](#)
[\[SN Utils\] Versions \(0\)](#)

Access Controls Choices (4) Attributes Labels (1)

Label Search

Choices

	Label	Value	Language	Sequence	Inactive	Updated	Dependent value
☐	Forgot Password	forgot password	en	2	false	2026-02-06 21:45:02	access
☐	Wi-Fi	wi-fi	en	0	false	2026-02-06 21:44:50	network
	Slow Computer	slow computer	en	3	false	2026-02-06 21:45:15	performance
	Projector	projector	en	1	false	2026-02-06 21:45:36	hardware

Insert a new row...

Incident Workflow
New record

Number INC00509

* Caller

Category Network

Subcategory -- None --

* Short description -- None --

Description

State -- None --

Assigned group

Assigned to

Submit

Incident WorkFlow

New record

Submit

Number

INC00509

* Caller

Category

Hardware

Subcategory

-- None --

* Short description

-- None --

Projector

Description

State

-- None --

Assigned group

Assigned to

Submit

Incident WorkFlow

New record

Submit

Number

INC00509

* Caller

Category

Access

Subcategory

-- None --

* Short description

-- None --

Forgot Password

Description

State

-- None --

Assigned group

Assigned to

Submit

Incident WorkFlow

New record

Submit

Number

INC00509

* Caller

Category

Performance

Subcategory

-- None --

* Short description

-- None --

Slow Computer

Description

State

-- None --

Assigned group

Assigned to

Submit

MILESTONE 3: Flow Designer Automation and Email Notification

For the automated part, the student only enters the Caller and Short Description. Category and Subcategory are filled by the flow. After the record is submitted, the flow checks the short description for keywords such as WiFi, Projector, Password, or Slow and assigns the matching values.

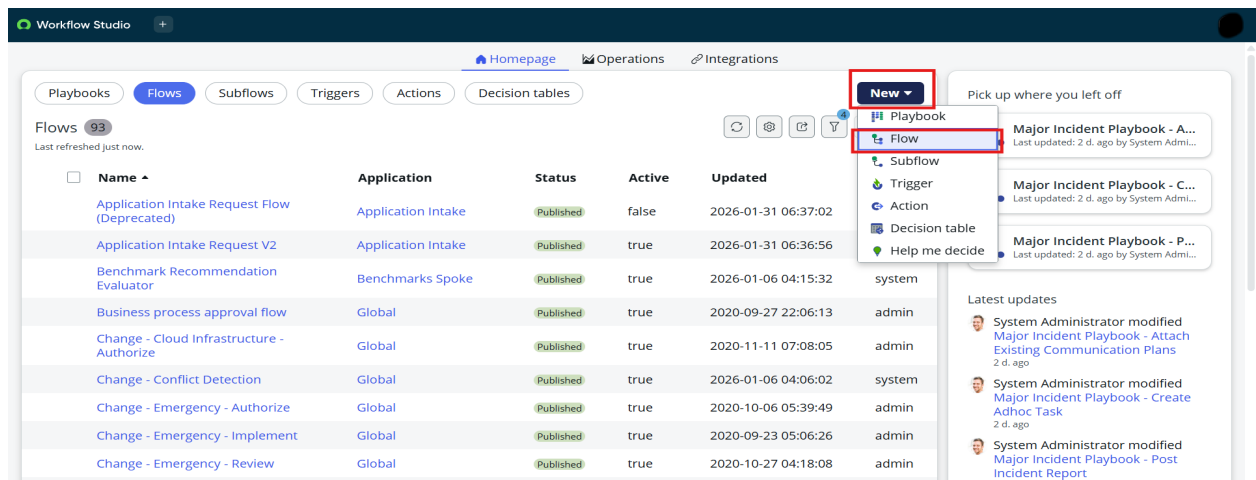
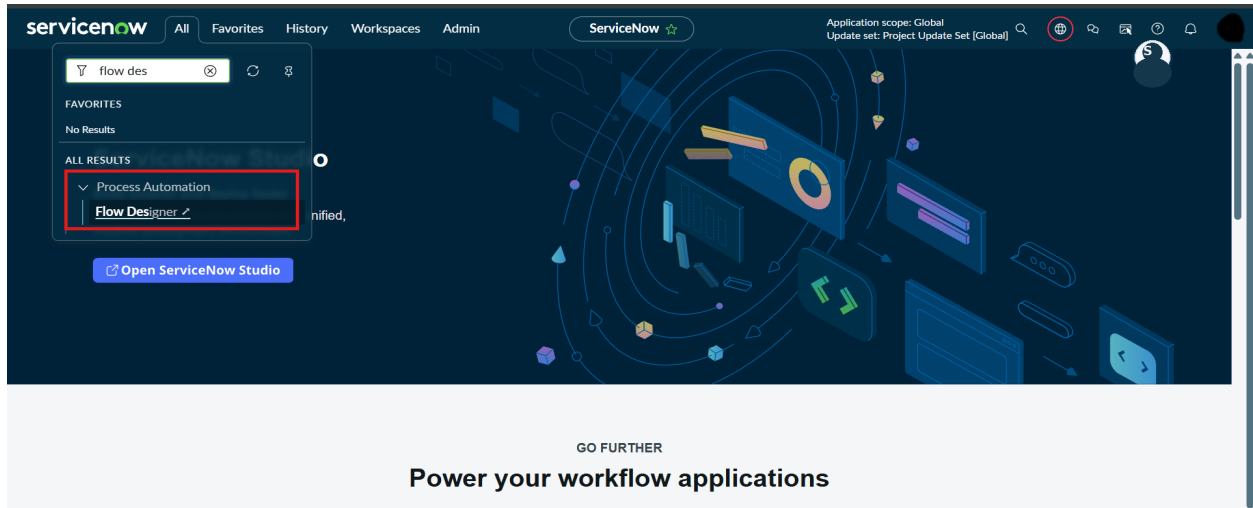
- If the short description contains WiFi → Category = Network, Subcategory = WiFi.
- If the description contains Projector → Category = Hardware, Subcategory = Projector.
- If the description contains Password or Login → Category = Access, Subcategory = Forgot Password.
- If the description mentions Slow or Hanging → Category = Performance, Subcategory = Slow Computer.

This makes classification more consistent and reduces the amount of manual checking required from the IT staff. The complete automation is built in Flow Designer, so it can be maintained and extended without scripts.

Activity 1: Auto Classification using Flow Designer

- Go to All menus → Search Flow designer.
- Under Process Automation, open Flow Designer.
- At the top right, click New and select Flow.
- Name: Auto Classify School IT Tickets.
- Application: Global.

- Now click the Build Flow button.



Let's get the details for your flow

Flow name *

Application *

Description

[Show additional properties](#)

Cancel
Build flow

Activity 2: Trigger Configuration

- Trigger: Record Created.
- Table: Incident Workflow.
- Condition: Category is Empty.
- Click Done.
- Under Actions, select the Flow Logic option.

Workflow Studio
Auto Classify School IT Tickets
Flow • Global

Auto Classify School IT Tickets
Inactive

View:
Test
Activate

TRIGGER

Incident WorkFlow Created where (Category is empty)

Trigger: Created

* Table: Incident WorkFlow [u_incident_w...

Condition: All of these conditions must be met

Category
is empty

New Criteria

Advanced Options

Delete
Cancel
Done

Data

Collapse All

Flow Variables

Trigger - Record Created

Incident WorkFlow Record
Record

Incident WorkFlow Table
Table

Run Start Time UTC
Date/Time

Run Start Date/Time
Date/Time

Activity 3: Wi-Fi Issue Classification

- Select the If condition.
- Condition label: SD contains Wi-Fi or Network.
- Condition 1: use the data pill picker to select Short description, operator Contains, value Wi-Fi.
- Condition 2: select Short description, operator Contains, value Network.
- Click Done.
- Click Action → Update record.
- In Record Field, drag the required data from the right side.
- Table: Incident Workflow.
- Set Category = Network.
- Set Subcategory = Wi-Fi.
- Click Done.

The screenshot shows the Workflow Studio interface for a workflow named "Auto Classify School IT Tickets". The workflow is currently inactive. The main workspace displays the configuration for an "If" condition (Step 1). The condition label is "SD is wifi OR network". There are two conditions defined:

- Condition 1: Trigger - ... Short descri... contains Wi-Fi
- Condition 2: Trigger - ... Short descri... contains Network

The conditions are connected by an "OR" operator. A red box highlights the "If" condition step and the "Done" button at the bottom right. The right sidebar shows the "Data" panel with a table of variables:

Variable	Type
Incident Workflow Record	Record
Incident Workflow Table	Table
Run Start Time UTC	Date/Time
Run Start Date/Time	Date/Time

Activity 4: Projector Issue Classification

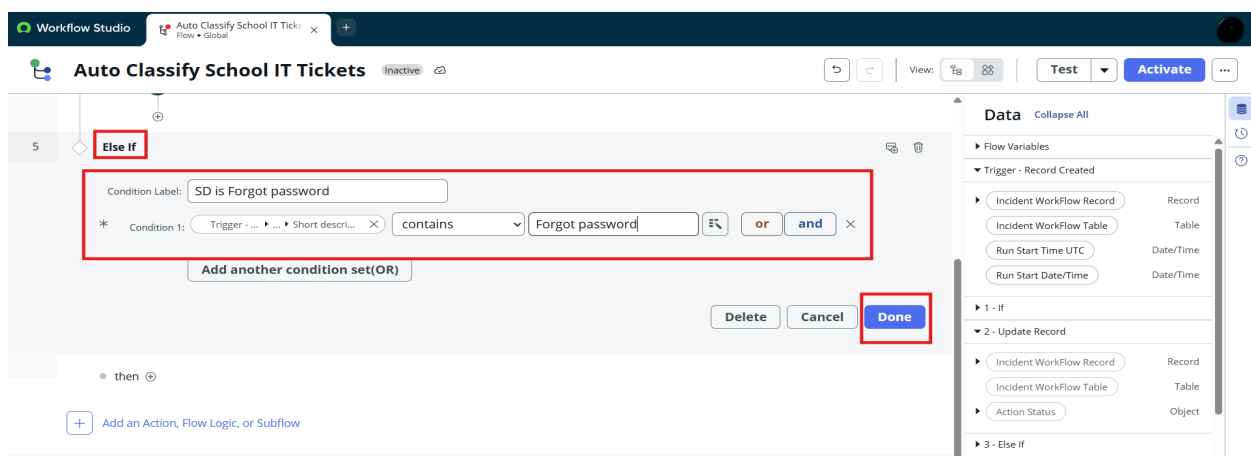
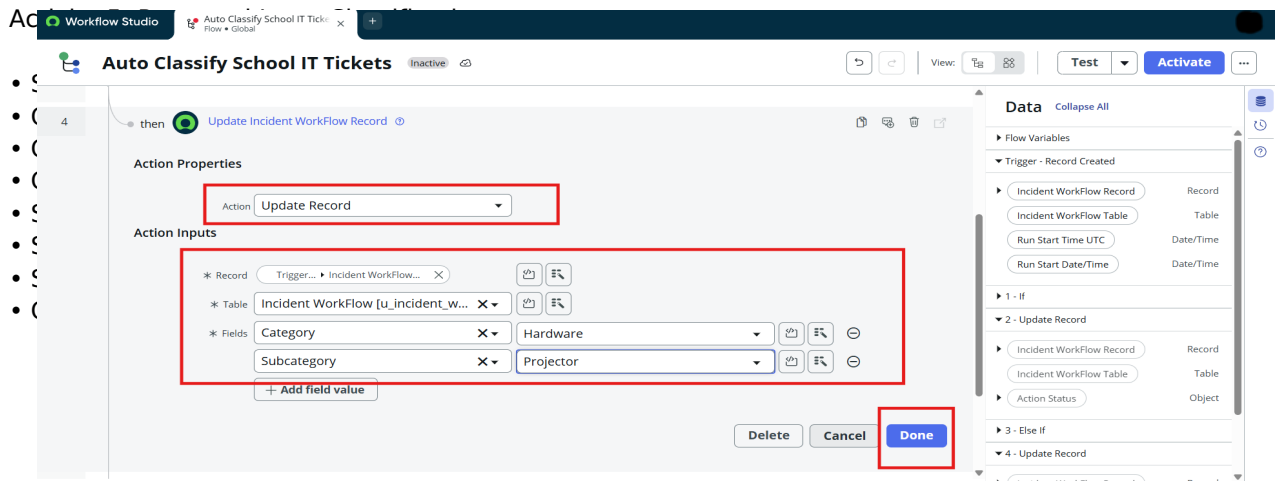
The screenshot shows the 'Auto Classify School IT Tickets' workflow in Workflow Studio. The workflow is currently 'Active'. Step 2 is 'Update Incident Workflow Record'. The 'Action Properties' section shows the action is 'Update Record'. The 'Action Inputs' section is highlighted with a red box and contains the following fields:

- * Record: Trigger... Incident Workflow...
- * Table: Incident Workflow [u_incident_w...]
- * Fields: Category (dropdown), Network (dropdown), Subcategory (dropdown), and Wi-Fi (dropdown).

Buttons for 'Delete', 'Cancel', and 'Done' are visible at the bottom right. The 'Data' panel on the right shows the flow variables, including 'Incident Workflow Record' and 'Incident Workflow Table'.

The screenshot shows the 'Auto Classify School IT Tickets' workflow in Workflow Studio. The workflow is currently 'Inactive'. Step 3 is 'Else If'. The 'Condition Label' is 'SD is projector OR hardware'. The 'Condition 1' is 'Trigger - ... Short descri...' contains 'Projector'. The 'or' and 'and' buttons are visible. The 'Done' button is highlighted with a red box. The 'Data' panel on the right shows the flow variables, including 'Incident Workflow Record' and 'Incident Workflow Table'.

- Click Done for the previous action.



Activity 6: Slow Computer Classification

The screenshot shows the Workflow Studio interface for the 'Auto Classify School IT Tickets' workflow. The workflow is currently inactive. The main canvas displays step 6, which is an 'Update Incident WorkFlow Record' action. The 'Action Properties' section shows the action is set to 'Update Record'. The 'Action Inputs' section is highlighted with a red box and contains the following fields:

- * Record: Trigger... Incident WorkFlow...
- * Table: Incident WorkFlow [u_incident_w...]
- * Fields: Category, Access, Subcategory, and Forgot Password.

Buttons for 'Delete', 'Cancel', and 'Done' are visible at the bottom right of the action configuration. A 'Data' panel on the right shows the flow variables, including 'Incident WorkFlow Record' (Record), 'Incident WorkFlow Table' (Table), 'Run Start Time UTC' (Date/Time), and 'Run Start Date/Time' (Date/Time).

The screenshot shows the Workflow Studio interface for the 'Auto Classify School IT Tickets' workflow. The workflow is currently inactive. The main canvas displays step 7, which is an 'Else If' condition. The 'Condition Label' is 'SD is Performance'. The 'Condition 1' is highlighted with a red box and contains the following fields:

- * Condition 1: Trigger - ... Short descri... contains Slow computer

Buttons for 'Delete', 'Cancel', and 'Done' are visible at the bottom right of the condition configuration. A 'Data' panel on the right shows the flow variables, including 'Incident WorkFlow Record' (Record), 'Incident WorkFlow Table' (Table), 'Run Start Time UTC' (Date/Time), and 'Run Start Date/Time' (Date/Time).

Activity 7: Email Notification Action

The screenshot shows the Workflow Studio interface for the 'Auto Classify School IT Tickets' workflow. The workflow is currently inactive. The configuration for step 8, 'Update Incident Workflow Record', is shown. The 'Action' is set to 'Update Record'. The 'Action Inputs' are configured as follows:

- * Record: Trigger... Incident Workflow...
- * Table: Incident Workflow [u_incident_w...]
- * Fields: Category (Performance), Subcategory (Slow Computer)

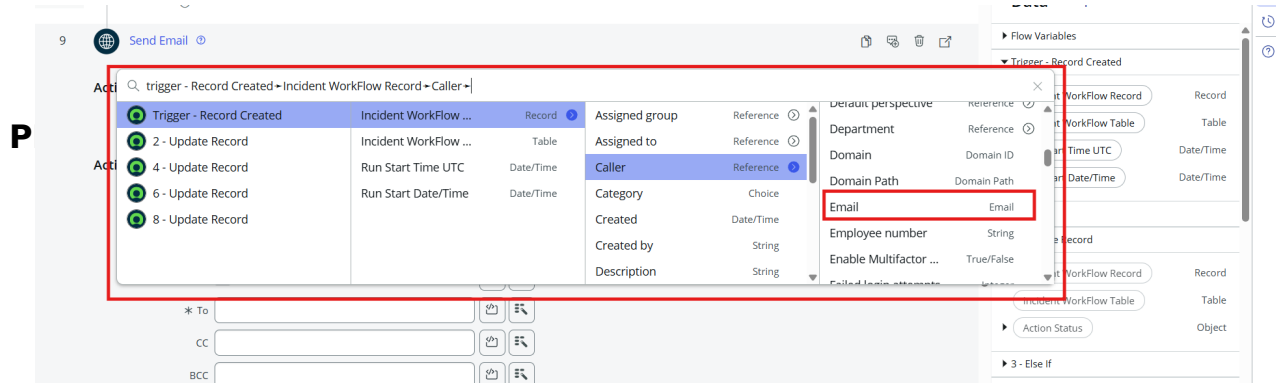
The 'Data' panel on the right shows the flow variables, including 'Incident Workflow Record' (Record), 'Incident Workflow Table' (Table), 'Run Start Time UTC' (Date/Time), and 'Run Start Date/Time' (Date/Time). The 'Done' button is highlighted.

The screenshot shows the Workflow Studio interface for the 'Auto Classify School IT Tickets' workflow. The workflow is now active. The configuration for step 9, 'Send Email', is shown. The 'Action' is set to 'Send Email'. The 'Action Inputs' are configured as follows:

- Target Record: Trigger... Incident Workflow...
- Table: Incident Workflow [u_incident_w...]
- Include Watermark: ☒
- * To: Trigger - Record C... Em...
- CC:
- BCC:
- * Subject: Your Request for the issue has been submitted.

The 'Data' panel on the right shows the flow variables, including 'Incident Workflow Record' (Record), 'Incident Workflow Table' (Table), 'Run Start Time UTC' (Date/Time), and 'Run Start Date/Time' (Date/Time). The 'Deactivate' button is highlighted.

- Click the Activate button from the top right corner.



Data Validation Checks

- Mandatory fields are captured correctly.
- Auto-number values are generated without duplication.
- Category and Subcategory are stored correctly.
- Reference fields resolve to the required user and group records.

Activity 1: Testing and Validation - Scenario 1

- Enter one Caller Name.
- Short Description: WiFi not working in library.
- Save the form.
- Reload the form.

- The flow fills the values automatically:
- Category → Network.
- Subcategory → Wi-Fi.
- An email is sent to the caller's email address.

A

Incident Workflow
INC00511

Number

* Caller

Category

Subcategory

* Short description

Description

State

Assigned group

Assigned to

Update Delete

servicenow

All Favorites History Workspaces Admin ServiceNow

Application scope: Global
Update set: Project Update Set [Global]

emails

FAVORITES
No Results

ALL RESULTS

System Logs

Emails

Bounce Email Addresses

Platform Analytics Administrat...

Schedules

Email Summaries

System Notification

GO FURTHER

Power your workflow applications

Activity 3: Test Scenario 2 - Projector Ticket

Emails Created Search Actions on selected rows... New						
All > Subject contains your						
Created	Recipients	Subject	Type	Notification type	User ID	
Search	Search	*your	Search	Search	Search	
2026-02-06 23:04:59	savannah.kesich@example.com	Your Request for the issue has been submitted.	send-ready	SMTP	(empty)	
2026-02-06 23:04:52	eliseo.wice@example.com	Your Request for the issue has been submitted.	send-ready	SMTP	(empty)	

Email

Your Request for the issue has been submitted

Subject

Recipients

Body

Content type

Headers

Preview Email

Hi greetings from xyz college!!!
We are viewed your request and solved by short time period.,
Thankyou ,
Ref:MSG0000396_ChBc6sAUD3zwGS0qlNyn

Update

Delete

Related Links

Preview Email

DSN:0005:VERSION:10/

Incident Workflow
INC00512

Number: INC00512

* Caller: Sam Sorokin

Category: Hardware

Subcategory: Projector

* Short description: Projector not turning on

Description:

State: -- None --

Assigned group:

Assigned to:

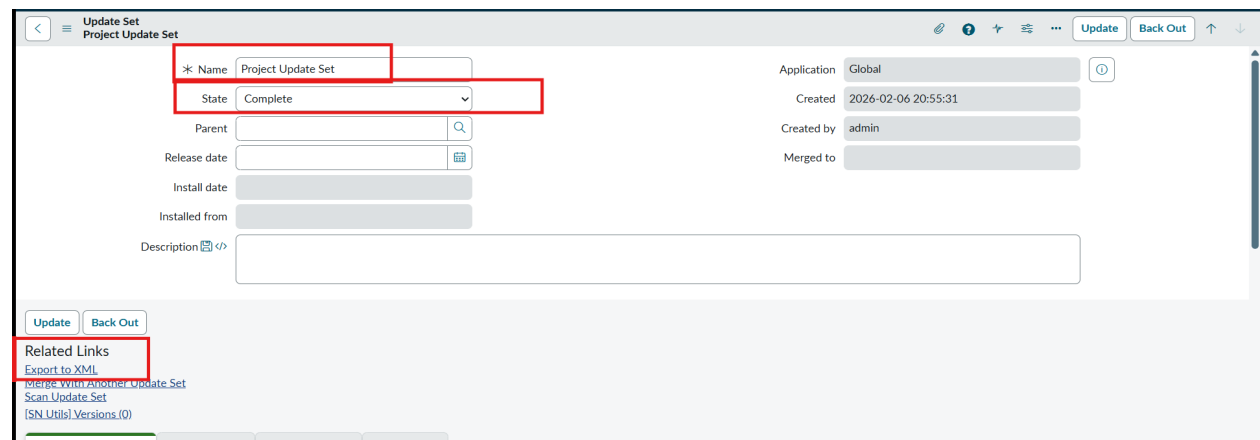
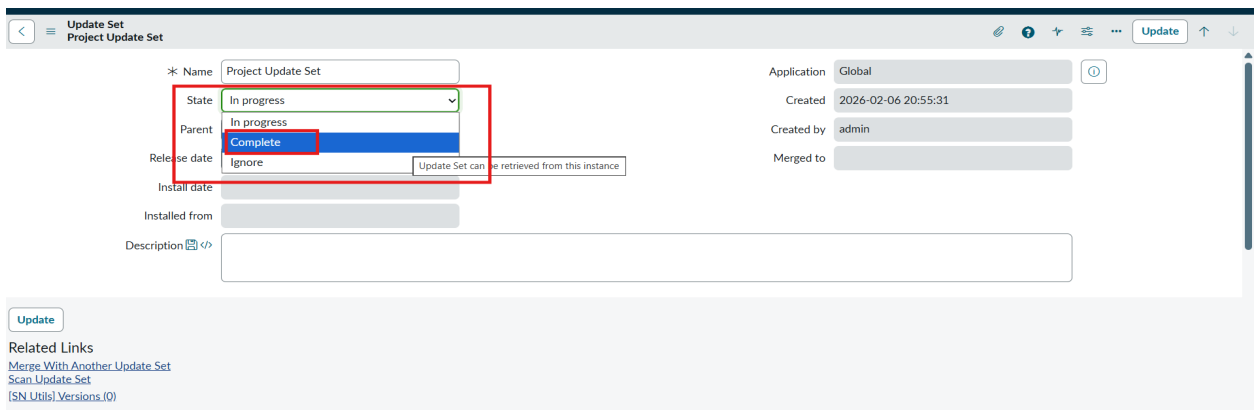
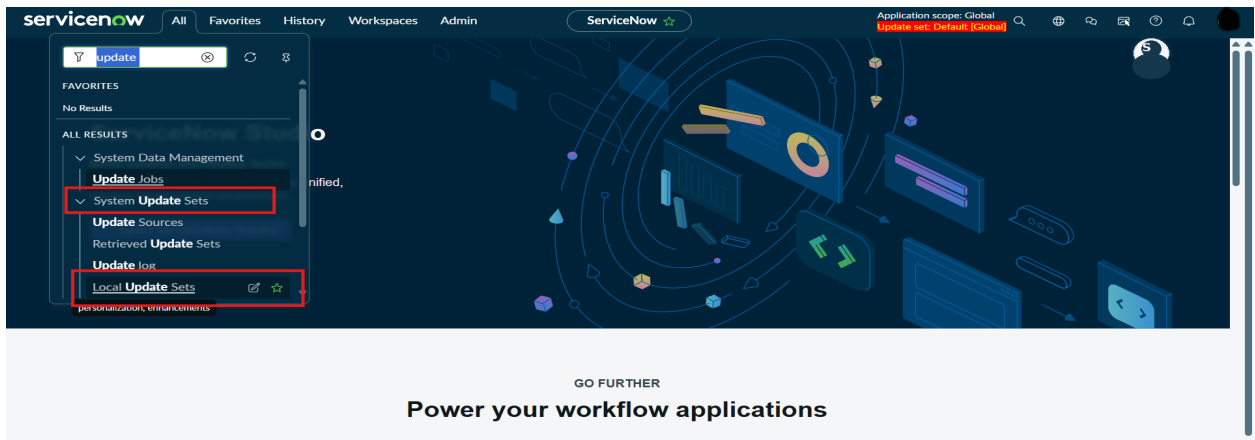
Update Delete

PHASE 5: Deployment and Final Documentation

The purpose of this phase is to move the completed update set to the required state and finish the project documentation. The solution is kept reusable and ready for demonstration or future use.

Activity 1: Moving the Update Set to Complete

- Go to All menus → Search for Local Update Sets.
- Under System Update Sets, select Local Update Sets.
- Open the Project Update Set.
- Change State from In progress to Complete.
- Save the form.
- Under Related Links, click Export to XML.
- The update-set data is downloaded in XML format.
- The downloaded file can be shared through email or another approved method.
- The update set can then be moved to another instance in the same way.



Browser address bar: d8%26sysparm_view%3D%26sysparm_domain%3Dnull...

Page header: atGPT Candidate Home

Application Update set:

Recent download history

	sys_remote_update_set_a7b40de6fc590ceaad36c.xml		
182 KB • Done			

Full download history 

ication Global

reated 2026-02-06 20:55:31

ited by admin

Conclusion / Final Notes

- This project documents an end-to-end ticket classification solution for a school IT helpdesk. The main aim was to reduce manual categorization work and make ticket handling more consistent.
- Using requirement analysis, backend configuration, dependent choice fields, Flow Designer, and email notification, the ticket can now be classified from the short description entered by the student.
- The project was completed in phases so that configuration, testing, and documentation could be checked step by step. The validation activities also checked the stored values and basic access controls.
- The final design is a no-code ServiceNow approach that can be maintained and extended later. Possible future additions mentioned in the project are assignment automation, SLA tracking, and predictive intelligence.